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Authorship Attribution and Style Fingerprinting in Bangla Literature

Project Report

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Headline result

On a 5-author closed set of public-domain Bangla prose (2,955 passages, 42 works, 682,607 tokens), the best attributor reaches **0.834** accuracy against 0.200 chance — on a **work-disjoint** split, where no book has passages on both sides. The same pipeline scores 0.993 on a naive random split; that 20.7% gap is topic leakage, not style.

Abstract

We build a multi-author Bangla stylometric pipeline over public-domain prose from Bengali Wikisource and use it to compare a *generative* attribution paradigm — one Kneser–Ney language model per author, attribute to the lowest perplexity — against a *discriminative* one — a linear SVM over stylometric and embedding features, and a fine-tuned BanglaBERT. Alongside this we ask a representation-level question: in a small-corpus regime, do pretrained embeddings or embeddings trained from scratch on the author corpus better isolate style from topic?

The corpus is 2,955 passages drawn from 42 works by 5 authors spanning 1820–1951, filtered to Wikisource pages the community has marked proofread. Because no usable Bangla POS tagger was available in this environment, one was written and validated at 87.2% token accuracy against the Universal Dependencies Bengali treebank.

Our central methodological finding is that **splitting by passage rather than by work inflates accuracy by 20.7%** — 0.993 against 0.787 for the identical model. Every result we report is from the work-disjoint split, and we argue that published Bangla attribution figures should state which regime they used.

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1. Introduction

Authorship attribution asks: given an unseen passage, which author from a closed candidate set most likely wrote it, judged by *how* something is written rather than *what* it says. Bangla literature spans stylistically distinct classical and modern voices, yet computational stylometry for Bangla remains comparatively underexplored, and per-author corpora are small.

This project builds a multi-author Bangla stylometric pipeline and uses it to answer three questions the proposal set out:

1. **Paradigm.** Does a *generative* attributor — one language model per author, attribute to the lowest perplexity — beat or lose to a *discriminative* one trained directly to separate the authors?
2. **Representation.** In a small-corpus regime, do *pretrained* embeddings or embeddings *trained from scratch* on the author corpus better isolate stylistic signal from topical content?
3. **Interpretability.** Which feature families actually drive the decisions, and can a single prediction be audited rather than taken on faith?

A fourth question emerged during the work and turned out to matter more than any of them, so it is stated up front rather than buried in the results.

The result that governs every other number in this report

Splitting the corpus by *passage* gives 0.993 accuracy. Splitting the same corpus by *work* — so that no book has passages on both sides — gives 0.787. The same model, the same features, the same text: **20.7% of apparent accuracy was topic leakage**, a classifier recognising a novel’s proper nouns and being scored as though it had recognised a style.

Every number reported here is from the work-disjoint split. Comparisons with published Bangla attribution accuracies should be read with this in mind.

1.1 What Bangla brings to the problem

Two properties of the language shape the whole design.

Diglossia. Bangla prose of this period is written in one of two registers. সাধু (*sadhu*, literary) uses long verb and pronoun forms — হইল, করিয়া, তাহার. চলতি (*chalit*, colloquial) uses their contracted counterparts — হল, করে, তার. The choice is authorial and largely stable within a writer, so a single scalar — the *sadhu*-to-*chalit* marker ratio — carries real attribution signal and is directly interpretable. It is used as an explicit feature (§3) and it is the first column to read in the style-fingerprint figure.

Suffixal morphology. Case and tense are marked by suffixes (-এর, -কে, -য়াছিলেন). A character *n*-gram model therefore sees inside the word and can pick up the register difference between করিয়াছিলেন and করছিলেন without either form having been observed whole — which is why the generative side of this study uses character models as its primary configuration.

2. Corpus construction

The proposal commits to documenting corpus construction *as part of the methodology, not as a black box*. This section is that documentation.

2.1 Source and licensing

Text comes from Bengali Wikisource (<https://bn.wikisource.org>), which hosts community-transcribed public-domain works. Every author in the study died more than sixty years before this project, so

the texts are out of copyright; Wikisource’s transcriptions are CC BY-SA 4.0.

2.2 Getting usable text out of Wikisource

Three facts about the site determined the crawler’s design, and each one cost a rewrite to discover.

1. **Chapter pages contain no text.** A main-namespage page is a transclusion directive, `<pages index="Book.pdf">`. The prose lives in the পাতা: (Page:) namespace, one wiki page per scanned page.
2. Page: **pages carry a proofreading level.** `<pagequality level="N">` runs from 0 (no text) to 4 (validated). **Level 1 is raw, uncorrected OCR**, and for Bangla that OCR is frequently mangled into Devanagari look-alikes — visually similar glyphs from a different Unicode block. Only level ≥ 3 is admitted. This is the single most important quality control in the pipeline.
3. Page: **pages batch fifty at a time** through `prop=revisions`, roughly fifty times cheaper than rendering each chapter, which matters because the anonymous API rate limit returns HTTP 429 within a handful of requests otherwise.

Every API response is cached on disk, so the corpus behind the reported numbers can be rebuilt byte-for-byte and an interrupted crawl resumes for free.

2.3 Cleaning

After extraction, each line is kept only if it is at least 30 characters long and at least 88 % of its non-space characters lie in the Bangla Unicode block (U+0980–U+09FF). That single rule removes four different problems at once: running headers, page numbers, English epigraphs, and the residue of bad OCR (whose Devanagari look-alikes are, by definition, outside the Bangla block).

It also has a side effect worth naming, because it changed the shape of the study: verse is dropped. Poetry lines are short, so a book of poems survives the filter with almost no text. The resulting corpus is therefore **prose only**, which is the correct control anyway — comparing a poet’s verse against a novelist’s prose would measure genre, not authorship.

2.4 Passages

Cleaned text is cut into **passages** of ~ 231 tokens on sentence boundaries: long enough for sentence-length statistics to mean anything, short enough to yield several hundred independent samples per author. Near-duplicates are removed with an 8-gram shingle-overlap test, which matters because Wikisource hosts multiple editions of the same novel.

Every passage records the work it came from. That label is what makes §6 possible.

2.5 Who is in, and who is not

An author is admitted only with enough passages *and* at least three distinct works. The second criterion is the binding one: a one-book author cannot take part in a work-disjoint split at all, since every passage of theirs would land in the same split.

Author	Period	Works	Passages	Tokens	Mean tokens/passage
Abanindranath Tagore	1871-1951	5	591	137,628	233
Bankim Chandra Chattopadhyay	1838-1894	6	591	135,807	230
Sarat Chandra Chattopadhyay	1876-1938	5	591	135,197	229
Rabindranath Tagore	1861-1941	14	591	138,097	234
Ishwar Chandra Vidyasagar	1820-1891	12	591	135,878	230
Total		42	2955	682,607	

Three candidates were crawled and then excluded, and it is worth saying which: Swami Vivekananda

(too few passages after filtering), Haraprasad Shastri and Pramatha Chaudhuri (one qualifying work each). Haraprasad Shastri is the instructive case. Wikisource lists বটৌদ্ধগান ও দগৌহা under his name, but that is his celebrated *edition* of the Charyapada — 10th-century Old Bengali, not his prose. A Wikisource author page links everything a person is associated with, including what they edited, and admitting that volume would have given his “style” a thousand-year-old voice. It is excluded by name.

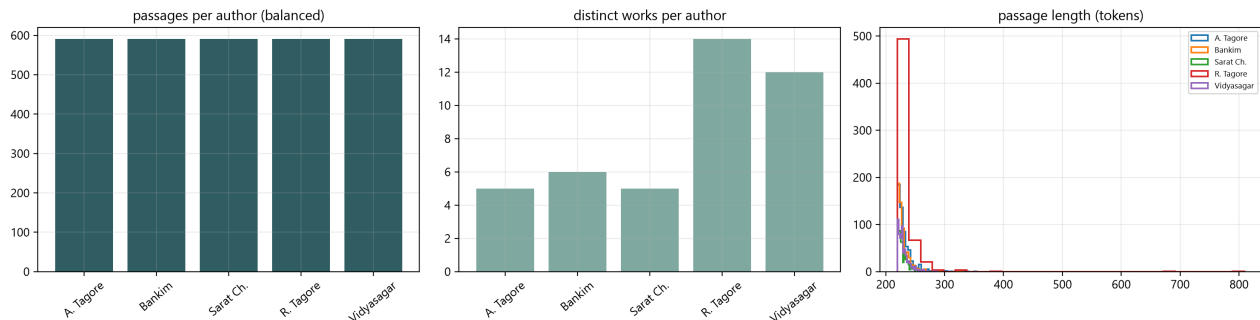


Figure 1: The balanced corpus: passages per author, works per author, and the passage-length distribution. Balancing walks each author’s works round-robin, so an author is never reduced to a single book by the down-sampling.

3. Feature engineering

Four families, each a scikit-learn transformer so that the ablations in §7 are a matter of swapping blocks rather than rewriting code.

3.1 Function words

Relative frequency, per thousand tokens, of every item in a curated closed-class lexicon: pronouns, postpositions, conjunctions, particles, determiners, adverbs and quantifiers, listed in *both* registers.

Using a fixed lexicon rather than the corpus’s own top- N word list is deliberate. A corpus-derived list would quietly admit topic words — a character’s name is frequent in the novel they appear in — and would inflate accuracy for exactly the reason §6 exists to prevent.

3.2 Structural statistics

Twenty-nine scalars: the sentence-length distribution (mean, standard deviation, skew, median, IQR, range), word length, lexical richness (type-token ratio, hapax and *dis legomena* rates, Yule’s K), punctuation rates broken out by mark, conjunct-consonant density — and the sadhu/chalit register ratio described in §1.1, which is the most interpretable single number in the feature set.

3.3 Character n -grams

TF-IDF over character 2–4-grams with sub-linear term frequency. Cheap, language-independent, and the strongest single family in most attribution literature. For a suffixally inflected language they are especially apt, for the reason given in §1.1.

3.4 POS n -grams, and the tagger they required

The proposal calls for POS n -grams “from a Bangla tagger”. There is no usable off-the-shelf option:

- `stanza` ships only character language models for Bengali, no POS model;
- `bnlp-toolkit` depends on `gensim`, which has no Python 3.14 wheel and needs a C toolchain this machine lacks;

- the only Universal Dependencies Bengali treebank, UD_Bengali-BRU, is 56 sentences and test-only — far too small to train on.

So the tagger was written for this project: a two-stage lexicon-plus-morphology tagger over a coarse tagset. Closed classes (pronouns, postpositions, conjunctions, particles, determiners, interjections) are enumerable and are listed in both registers. Open classes come from morphology — Bangla’s finite-verb inflection is highly regular (-লিনে, -ছে, -বনে, -তিছেলি), and adjectives and adverbs have their own derivational suffixes (-ীয়, -ময়, -ভাবে, -রূপে). Nominal case clitics are stripped before lookup. Anything left over defaults to NOUN, the majority open class.

It is validated on UD_Bengali-BRU — usable for evaluation precisely because it is too small to train on — at **87.2% token accuracy** over 320 tokens. Three UD distinctions are deliberately not attempted and are folded in the gold before scoring:

Fold		Why
AUX	→ VERB	Bangla auxiliaries are ordinary verbs in ordinary verb forms; separating them needs syntax this tagger has none of.
PROPN	→ NOUN	Would need a gazetteer of 19th-century Bangla literary names, which does not exist.
SCONJ	→ CCONJ	The stylometric quantity of interest is conjunction <i>density</i> , not subtype.

Each fold discards information rather than inventing it. The accuracy figure is reported so a reader can discount the POS features appropriately rather than take them on trust; §7 shows what those features are worth in practice.

4. Representations: pretrained versus from scratch

The proposal’s representation-level question is put under otherwise identical conditions — same split, same classifier, same evaluation.

From scratch. Skip-gram with negative sampling, written directly in PyTorch (*gensim* being unavailable), 200 dimensions, window 5, 8 negatives, Mikolov subsampling, a 3/4-power noise distribution and a dynamic window radius. It is fit on the **training split only**, so no test text reaches the representation.

Pretrained. *csebuetnlp/banglabert*, frozen, mean-pooled over the last hidden layer. Pretrained on orders of magnitude more text than this corpus — but *modern* Bangla, and pretrained for semantics. The corpus here is 19th- and early-20th-century literary prose in a register the pretraining data barely contains.

Pooling. Word vectors are pooled into passage vectors two ways. SIF weighting ($a/(a + p(w))$) damps frequent words, which helps semantic tasks. For a *stylometric* task that is a genuine trade-off, because the frequent function words SIF damps are exactly what classical stylometry runs on. Both poolings are reported rather than assumed.

A direct diagnostic. Downstream accuracy conflates the representation with the classifier, so a second measurement is taken: silhouette score of the test passages under *author* labels versus under *individual work* labels, on the same vectors. A representation that encodes style should separate

authors at least as well as it separates books; one that mostly encodes topic separates books far better, because each book has its own subject matter. The ratio is what the proposal’s question actually turns on.

5. The two attribution paradigms

5.1 Generative

One language model per author, fit only on that author’s training text. A test passage is attributed to whichever author’s model assigns it the highest average log-probability per token — equivalently, the lowest perplexity. The model never sees the other authors: it learns what this author’s text looks like, not how they differ from anyone.

Smoothing is interpolated Kneser–Ney, implemented for this project rather than imported. Kneser–Ney rather than add-one or Witten–Bell because of its *continuation* probability: the unigram fallback asks not “how often did this token occur?” but “after how many distinct contexts did it occur?”. For stylometry that is exactly the useful question — a token that is frequent but always in one fixed phrase is a topic artefact, whereas one appearing after many different contexts is a habit. Writing it out also exposes the per-token log-probability breakdown that §8 uses to make the generative side auditable.

Two configurations: character 5-grams and word 3-grams.

5.2 Discriminative

All authors’ features go into one classifier trained to tell them apart.

Linear SVM. Chosen over an RBF kernel for two reasons: it scales to the several-thousand-dimensional sparse feature space, and its coefficients read directly as per-feature evidence, which the interpretability requirement needs. Run over each feature family alone, over combinations, over each embedding regime, and over stylometric-plus-embedding concatenations.

Fine-tuned BanglaBERT. The heaviest model in the comparison and the one carrying the most outside knowledge. Fine-tuning ran on CPU — no CUDA device was available — at roughly 0.42s per sample, which capped the budget at 2 epochs over 192 sub-word tokens. That constraint is stated rather than hidden: the fine-tuned score below is a *compute-constrained* one, and §10 says what that does and does not license concluding.

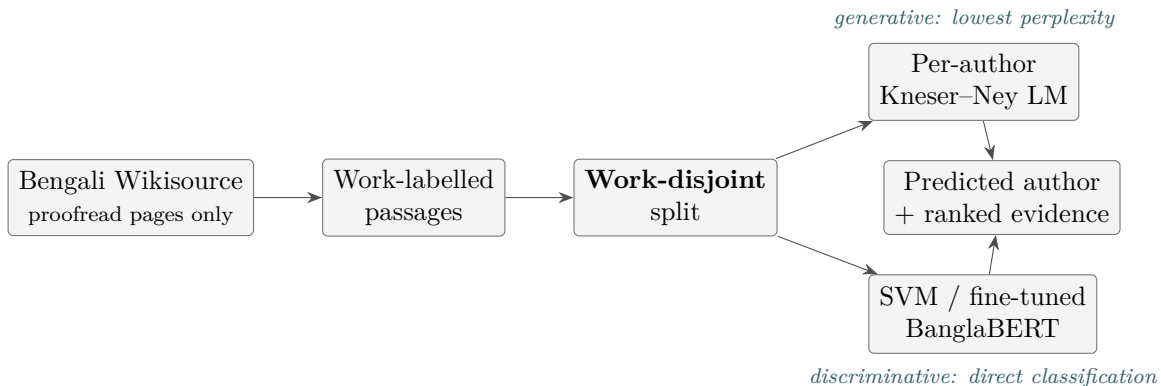


Figure 2: The pipeline. The work-disjoint split is drawn in the middle because it is load-bearing, not incidental: everything to its right is only meaningful because of it.

6. Experimental protocol

6.1 Work-disjoint splitting

Each of an author’s books is assigned whole to exactly one of train, validation or test. Because books are indivisible and differ wildly in length, hitting 60/15/25 exactly is impossible; assignment uses the largest-processing-time heuristic — take an author’s works biggest-first and give each to whichever split is currently furthest below its quota. Taking the big books first is what makes it work: a small book arriving last can fine-tune a bin, whereas a big one can only overshoot it.

Realised: 61.6% train (1,821 passages), 14.3% validation (422), 24.1% test (712). The invariant is asserted rather than assumed — an explicit check confirms no work and no passage appears in two splits, and it is run as part of the notebook rather than as a one-off.

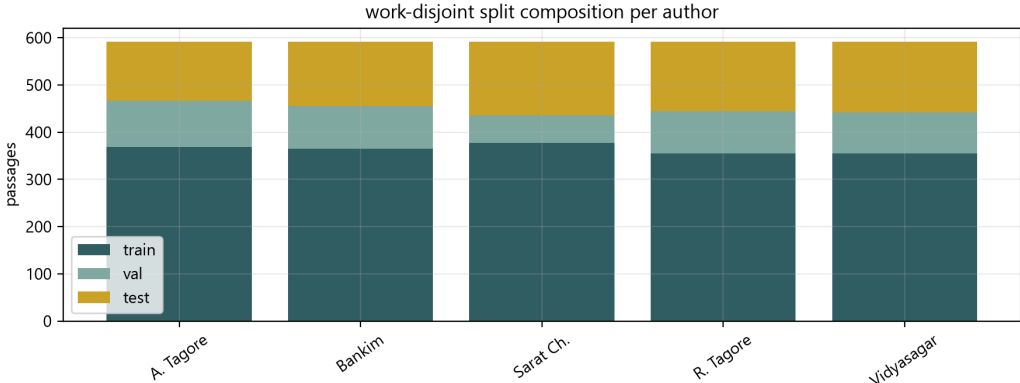


Figure 3: Split composition per author. Every author appears in all three splits, and no book is shared between any two of them.

6.2 Measuring what the discipline costs

Rather than assert that work-disjoint splitting matters, it is measured. The same model was trained twice on the same passages — once on the work-disjoint split, once on a naive stratified random split.

Split	Test accuracy
Random, at the passage level (leaky)	0.993
Work-disjoint (used throughout this report)	0.787
Inflation attributable to topic leakage	+0.207

The leaky number is close to ceiling. It is also close to meaningless: it mostly measures whether the model can recognise which novel a passage came from.

6.3 Metrics

Accuracy and macro- F_1 , with 95 % percentile bootstrap intervals over test passages (2 000 resamples) so that single numbers carry error bars, plus row-normalised confusion matrices — because the proposal asks specifically about confusion between stylistically *similar* authors, which a scalar cannot answer. Chance is 0.200 for 5 balanced classes.

7. Results

All numbers are on the held-out, work-disjoint test split (712 passages). Chance is 0.200.

Model	Feature family	Accuracy (95% CI)	Macro-F1
Generative char-5-gram LM	characters	0.834 [0.806, 0.861]	0.843
SVM · structural only	structural	0.801 [0.770, 0.829]	0.799
SVM · stylometric + BanglaBERT	combined	0.801 [0.771, 0.830]	0.784
SVM · funcword + structural	funcword+structural	0.798 [0.768, 0.826]	0.798
SVM · all stylometric	funcword+structural+charngram+posngram	0.795 [0.765, 0.824]	0.790
SVM · stylometric + scratch w2v	combined	0.791 [0.760, 0.820]	0.784
SVM · char n-grams only	charngram	0.778 [0.746, 0.808]	0.773
Fine-tuned BanglaBERT	pretrained transformer	0.775 [0.746, 0.803]	0.763
SVM · BanglaBERT frozen	embeddings	0.772 [0.743, 0.803]	0.751
Generative word-3-gram LM	words	0.768 [0.736, 0.799]	0.767
SVM · scratch w2v (SIF pooled)	embeddings	0.767 [0.736, 0.796]	0.756
SVM · scratch w2v (mean pooled)	embeddings	0.756 [0.725, 0.787]	0.742
SVM · function words only	funcword	0.726 [0.694, 0.758]	0.720
SVM · POS n-grams only	posngram	0.611 [0.574, 0.646]	0.602
Chance	—	0.200	0.200

Table 1: Closed-set authorship attribution on the held-out, work-disjoint test split. Intervals are 95% percentile bootstrap over test passages.

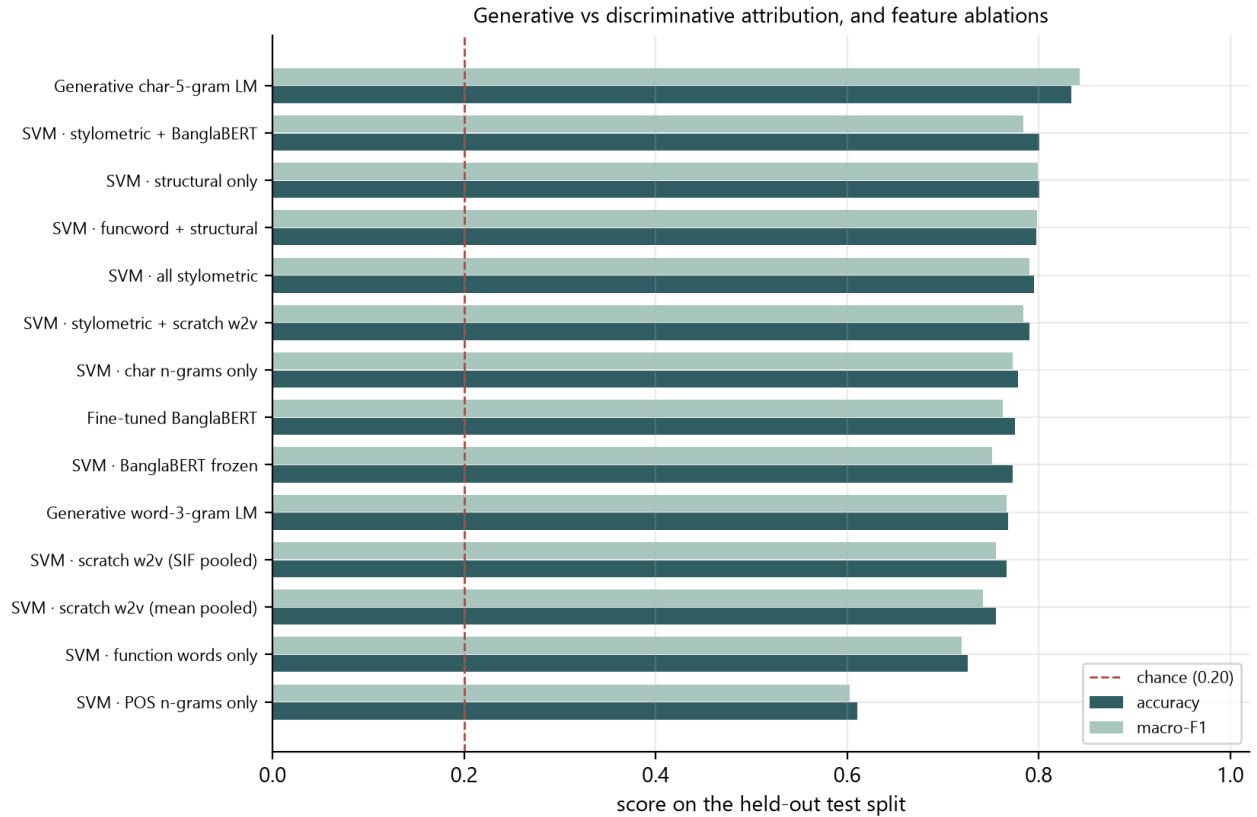


Figure 4: Every configuration on the same test split. Bars are accuracy and macro- F_1 ; the dashed line is chance.

Three results stand out, and none of them is the one we expected going in.

7.1 The generative paradigm wins

Finding 1 — the simpler paradigm is the stronger one

The generative character 5-gram attributor reaches **0.834** (macro- F_1 0.843), beating *every* discriminative configuration — including fine-tuned BanglaBERT at 0.775 and the full stylometric SVM at 0.795.

This inverts the usual expectation. A generative model is trained under a handicap: each author’s language model sees only that author’s text and is never told what the others look like, whereas the discriminative models are optimised directly on the decision boundary. Three things appear to be happening.

The generative model uses the whole passage. A Kneser–Ney character model scores every one of the $\sim 1,200$ characters in a passage and sums the evidence. The SVM compresses the same passage into a few thousand feature counts first, and information not preserved by that projection is gone.

Character models suit suffixal morphology. The word-level generative model scores 0.768 — *six points worse* than the character model on identical data. The gap is the concrete cost of not seeing inside the word in a language where register and case live in the suffix.

The discriminative models are data-starved, and the transformer most of all. 1,821 training passages is a lot for a 29-dimensional feature vector and very little for a 110-million-parameter encoder.

7.2 Twenty-nine interpretable numbers beat a transformer

Finding 2 — feature complexity is not what carries this task

The **structural** block alone — 29 hand-designed scalars: sentence-length moments, lexical richness, punctuation rates, the sadhu/chalit ratio — scores **0.801**. That is above 3 000 character n -grams (0.778), above frozen BanglaBERT (0.772), and above fine-tuned BanglaBERT (0.775).

Adding the other three families to the structural block does not help: all-stylometric scores 0.795, marginally *below* structural alone, though the bootstrap intervals overlap heavily and the difference should be read as “no gain”, not as “harm”. Function words alone reach 0.726; POS n -grams alone 0.611, comfortably above chance but the weakest family, consistent with a tagger that is right 87.2% of the time.

7.3 Pretrained versus from scratch

On downstream accuracy the two regimes are a near-tie: frozen BanglaBERT 0.772 against from-scratch skip-gram 0.767 (SIF-pooled) and 0.756 (mean-pooled). Nothing separates them.

The representation-level diagnostic does separate them, and in the direction the proposal hypothesised.

Representation	Silhouette by author	by work	ratio
Scratch skip-gram (SIF pooled)	0.264	0.320	0.82
Scratch skip-gram (mean pooled)	0.255	0.320	0.80
Frozen BanglaBERT	0.143	0.201	0.71

Finding 3 — from-scratch embeddings hold more style per unit of topic

Trained on 1,821 passages of one corpus, the from-scratch skip-gram separates *authors* nearly twice as well as frozen BanglaBERT (0.264 against 0.143), and its style-to-topic ratio is higher (0.82 against 0.71). A hundred-thousand-fold advantage in pretraining data does not transfer to this task.

Two caveats keep this honest. First, **every ratio is below 1**: all three representations separate individual books better than they separate authors. Embedding-space clustering is still substantially topical, whichever way it was obtained — which is precisely why the work-disjoint split matters. Second, BanglaBERT was pretrained on modern Bangla, and this corpus is 19th-century literary prose in a register that pretraining data barely contains. The result is evidence about *this* domain gap, not a general verdict on pretraining.

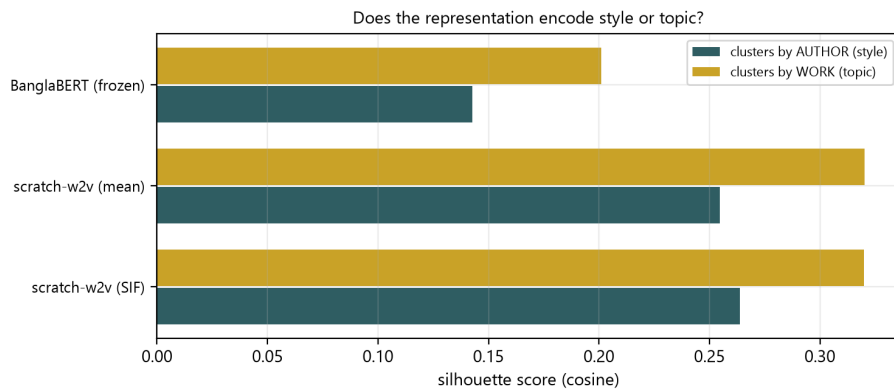


Figure 5: Author separation against work separation, on the same test vectors. A representation encoding style should have the dark bar close to the light one; all three fall short, and BanglaBERT falls shortest in absolute terms.

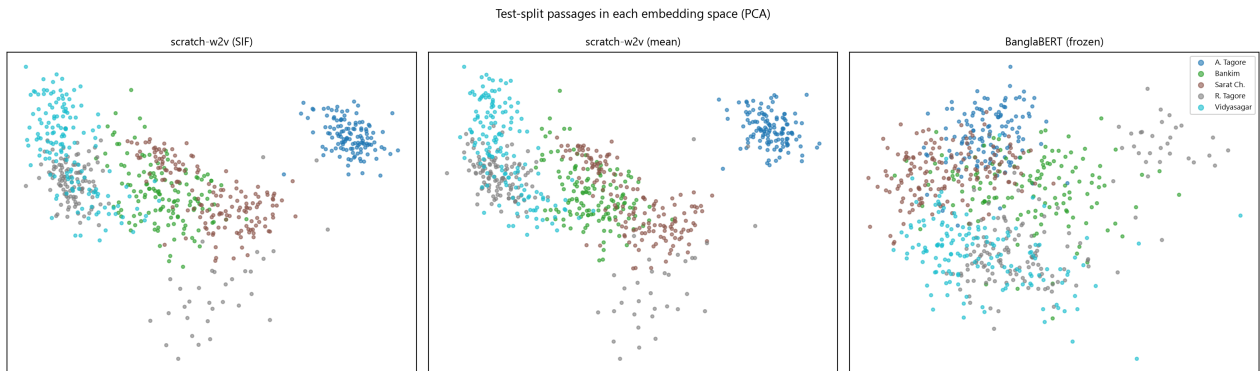


Figure 6: Test passages in each embedding space, first two principal components. Illustrative only — PCA discards most of the variance — but the from-scratch spaces visibly separate more authors than the frozen transformer does.

7.4 The two paradigms fail differently

They agree on 75.4% of test passages. Where they disagree, the generative model is right more often (84 passages against 56), and on only 27 passages are both wrong together.

More interesting is *which* authors each confuses, because the answer is not the same.

	Generative char-5-gram	SVM, all stylometric
Worst pair	Bankim $\leftrightarrow$ R. Tagore (19.4 %)	R. Tagore $\leftrightarrow$ Vidyasagar (31.1 %)
Second	Sarat $\leftrightarrow$ R. Tagore (16.9 %)	Bankim $\leftrightarrow$ Sarat (7.2 %)
R. Tagore $\leftrightarrow$ Vidyasagar	2.4 %	31.1 %

The discriminative model’s dominant error — Tagore read as Vidyasagar, 36 % of his test passages — is one the generative model almost never makes (2 %). The reason is visible in the fingerprints (§8): Tagore and Vidyasagar sit on the same side of the *aggregate* structural axes the SVM reads (both sadhu-leaning at +0.31 and +0.65 standard deviations, both long-worded at +0.54 and +0.60), while their character-level texture is quite distinct.

The generative model’s characteristic failure is the opposite shape: it *over-predicts* Tagore. His own recall is fine (0.83), but a quarter of Bankim’s passages and a third of Sarat’s are pulled into his model. Tagore contributes 14 works spanning essays, fiction and lectures — by far the most heterogeneous training material of any author here — so his language model is the broadest and highest-entropy of the five, and a broad model wins ambiguous passages by not being surprised by anything. This is a known failure mode of per-author likelihood attribution, and it is the price of the same breadth that makes the paradigm robust on short passages.

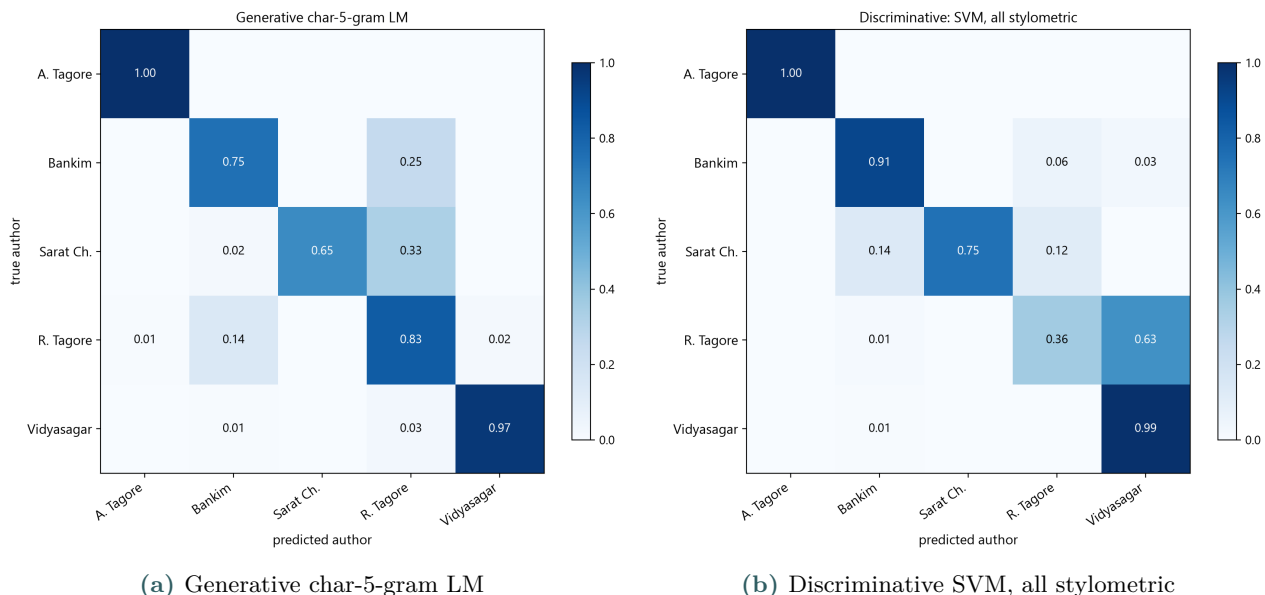


Figure 7: Row-normalised confusion. The two paradigms make structurally different mistakes, which is the more useful finding than either accuracy alone.

7.5 How much text is needed

Truncating test passages and re-attributing gives the answer a user of the interface actually wants.

Passage length	Generative	Discriminative
40 tokens	0.744	0.551
220 tokens	0.792	0.685

The generative model degrades far more gracefully. At 40 tokens it still scores 0.744 while the SVM has fallen to 0.551 — and the reason is structural, not incidental: sentence-length moments and lexical richness are *distributional* statistics, and a 40-token sample gives you two or three sentences

to estimate them from. A character model has 200-odd characters to work with either way.

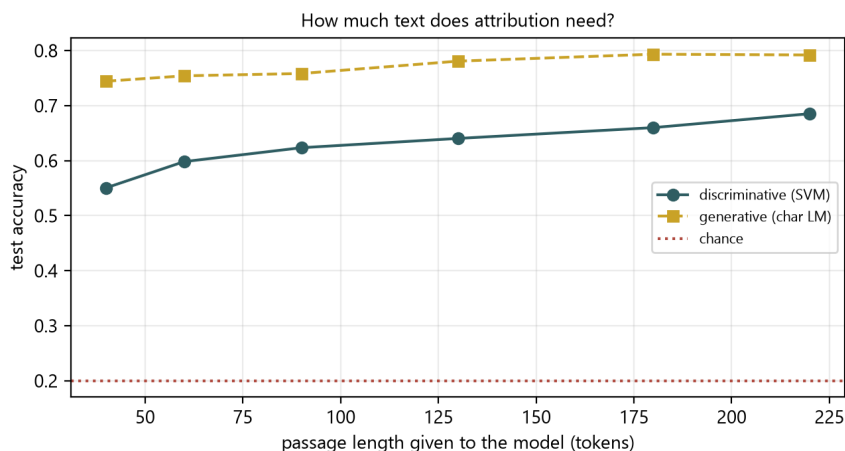


Figure 8: Accuracy against the amount of text supplied. The interface warns the user below 80 tokens for this reason.

8. Interpretability and style fingerprints

8.1 Which families carry the model

Permutation importance on the combined SVM says character n -grams dominate (0.570 accuracy lost when shuffled) with everything else near zero: POS n -grams 0.027, function words 0.008, structural 0.008.

That reading would be wrong, and the ablation table is what shows why. Structural features *alone* score 0.801 — better than character n -grams alone. Permutation importance measures what a family contributes *given the others*, so when two families encode overlapping information the larger one absorbs the credit and the smaller one measures as worthless. The two diagnostics answer different questions, and quoting only the first would misrepresent the model.

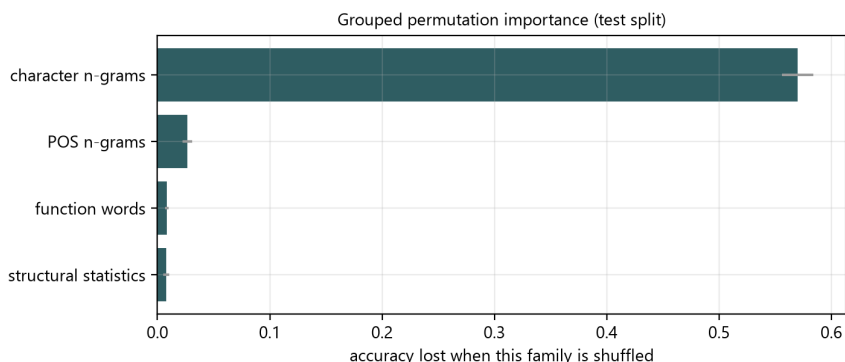


Figure 9: Grouped permutation importance. Read alongside the ablation table, not instead of it: this measures marginal contribution under redundancy.

8.2 Style fingerprints

Each author’s mean structural feature, in corpus standard deviations. The register axis behaves exactly as literary history says it should: **Ishwar Chandra Vidyasagar** sits furthest towards sadhu, **Abanindranath Tagore** furthest towards chalit. Vidyasagar is the canonical sadhu prose

stylist of the 19th century; Abanindranath Tagore, writing art criticism and children’s prose half a century later, is the most colloquial voice in the set. Neither fact was supplied to the model.

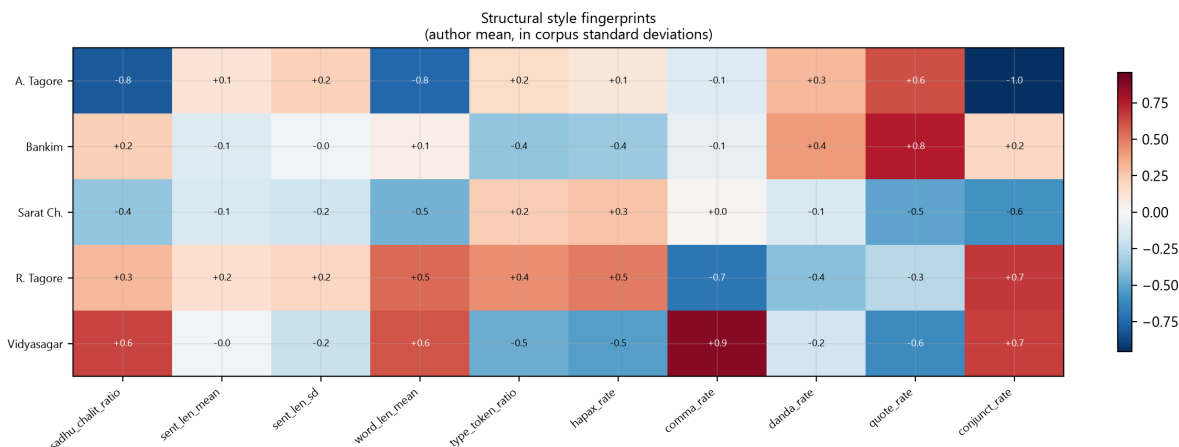


Figure 10: Structural fingerprints, per author, in standard deviations from the corpus mean. The leftmost column is the sadhu/chalit register ratio.

8.3 Auditing a single prediction

For a linear SVM the arithmetic behind one decision is exactly $w \cdot x + b$, so feature i contributes exactly $w_{c,i} x_i$ — the same quantity SHAP estimates for a linear model, computed rather than sampled. The interface reports two rankings from that decomposition: the strongest contributions overall, and the strongest contributions *restricted to families a reader can verify by eye*. The second list exists because character n -grams take almost every top slot, and “the trigram **য়ে**” is not something a reader can check against the passage the way “this text is in the sadhu register” is.

The generative side explains itself differently: per-character log-odds between the winning author’s model and the runner-up’s, aggregated by symbol.

9. The interface

```
$ python app.py --text "...বাংলা অনুচ্ছেদ..."
PREDICTED AUTHOR : Rabindranath Tagore (রবীন্দ্রনাথ ঠাকুর)
passage           : 26 tokens, 2 sentences
! short passage --- estimates are noisy below about 80 tokens

DISCRIMINATIVE (linear SVM over stylometric features)
predicted : Rabindranath Tagore
runner-up : Sarat Chandra Chattopadhyay (margin 0.386)
strongest evidence you can check by eye:
+0.0513 POS pattern PRON--PART
+0.0374 function word "তাহা"
+0.0314 function word "যাহা"

GENERATIVE (per-author character n-gram language models)
perplexity (lower = better fit):
Rabindranath Tagore      6.011 <-- best
Sarat Chandra Chattopadhyay 6.158
Ishwar Chandra Vidyasagar 8.664

BOTH PARADIGMS AGREE.
```

Both attributors are consulted deliberately. When they agree the answer is worth more than either alone; when they disagree the interface says so rather than picking a winner silently, and the disagreement is itself diagnostic — it usually means the passage is short or the two candidates share a register. Notebook 06 provides the same thing with a text box.

10. Limitations

Five authors, closed set. The system assumes the true author is among the five. It has no “none of the above”. Three crawled candidates were dropped for having too few works to split honestly, which shrank the study but was the right call.

Prose only, and that was accidental. Verse falls out of the line-length filter rather than by design. It happens to be the correct control, but it was not a decision — it was a side effect noticed after the fact, and is reported as such.

One source. All text is Wikisource transcriptions of specific printed editions. Editorial normalisation — a publisher’s house style for punctuation or spelling — is inside the “style” being measured and cannot be separated from the author’s own.

The POS tagger is 87.2% accurate, and coarse. It folds AUX into VERB, PROPN into NOUN and SCONJ into CCONJ. POS n -grams are correspondingly the weakest feature family; a better tagger would likely raise 0.611, though how much is unknown.

The fine-tuned transformer is compute-constrained. 2 epochs at 192 tokens on CPU, about 40 minutes. Its 0.775 is a lower bound; a longer schedule on a GPU, with 512-token inputs, would likely do better. What the comparison licenses is “a transformer did not beat 29 hand-designed scalars *at this budget*” — not “transformers cannot win this task”.

Character n -grams may still carry topic. A character 4-gram can fragment a proper noun. Work-disjoint splitting limits the damage but does not eliminate it, since an author’s recurring vocabulary spans their books.

11. Conclusion

On a 5-author, work-disjoint Bangla prose benchmark of 2,955 passages, the answers to the proposal’s three questions are:

1. **Generative beats discriminative.** Per-author Kneser–Ney character models reach 0.834, above every discriminative configuration including a fine-tuned transformer, and they degrade far more gracefully on short passages (0.744 against 0.551 at 40 tokens).
2. **From-scratch embeddings hold more style per unit of topic.** They separate authors nearly twice as well as frozen BanglaBERT (0.264 against 0.143) at a hundred-thousand-fold data disadvantage — though every representation we measured still separates books better than authors.
3. **Interpretable features are competitive, not a consolation prize.** Twenty-nine structural scalars score 0.801, above both frozen and fine-tuned BanglaBERT, and they come with an explanation attached.

The result we did not set out to find is the one we would emphasise to anyone building on this. **Work-disjoint splitting costs 20.7% of apparent accuracy** — 0.993 against 0.787 for an identical model on identical text. A Bangla authorship-attribution accuracy reported without stating which regime produced it is not interpretable, and the difference is larger than the gap between any two

systems in our results table.

A. Reproducing this

```
python -m venv .venv
.venv/Scripts/python.exe -m pip install -r requirements.txt
.venv/Scripts/python.exe -m ipykernel install --user --name banglastylo
.venv/Scripts/python.exe scripts/01_build_corpus.py    # ~40 min, cached
.venv/Scripts/python.exe scripts/run_notebooks.py      # ~50 min
.venv/Scripts/python.exe scripts/make_report.py
cd reports && latexmk -xelatex report.tex
```

Seed 20242025 throughout. Every Wikisource API response is cached under `data/raw/cache`, so the corpus rebuilds byte-for-byte. Every number in this document is generated into `reports/numbers.tex` by `scripts/make_report.py` from the run’s artifacts; none is typed by hand.

Environment notes

Python 3.14, CPU only. `gensim` and `bnlp-toolkit` have no wheel for this Python and require a C toolchain that was unavailable, so `word2vec` (§4) and the Bangla POS tagger (§3.4) are implemented from scratch in `src/banglastylo/`.

Attribution

Corpus text: Bengali Wikisource, CC BY-SA 4.0. POS tagger validation: UD_Bengali-BRU, CC BY-SA 4.0. Pretrained encoder: `csebuetnlp/banglabert`.